BLOCKCHAIN TECHNOLOGY IN DECENTRALIZED FINANCE (DEFI)

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INTRODUCTION

TRADITIONAL FINANCE SYSTEMS LIKE BANKS AND FINANCIAL INSTITUTIONS DEPEND ON INTERMEDIARIES TO PROCESS TRANSACTIONS.

THIS CAUSES PROBLEMS SUCH AS:

- HIGH TRANSACTION FEES
- SLOW PROCESSING
- LOW TRANSPARENCY
- LIMITED ACCESS FOR MANY PEOPLE
- BLOCKCHAIN TECHNOLOGY PROVIDES A NEW SOLUTION BY CREATING A **DECENTRALIZED SYSTEM** WHERE TRANSACTIONS HAPPEN DIRECTLY BETWEEN USERS. THIS HAS LED TO **DECENTRALIZED FINANCE (DEFI)**, WHERE SERVICES LIKE LENDING, BORROWING, TRADING, AND PAYMENTS WORK WITHOUT BANKS OR MIDDLEMEN.

WHAT IS DEFI

- DEFI MEANS **DECENTRALIZED FINANCE**.
  IT IS A FINANCIAL SYSTEM BUILT ON BLOCKCHAIN USING **SMART CONTRACTS**.

MAIN FEATURES:

- NO INTERMEDIARIES
- PEER-TO-PEER TRANSACTIONS
- AUTOMATED SYSTEMS
- TRANSPARENT OPERATIONS
- GLOBAL ACCESSIBILITY

POPULAR DEFI PLATFORMS:

- AAVE
- UNISWAP
- MAKERDAO
- THESE PLATFORMS RUN MAINLY ON THE ETHEREUM BLOCKCHAIN

BLOCKCHAIN IN DEFI

BLOCKCHAIN WORKS AS A DISTRIBUTED DIGITAL LEDGER.
EVERY TRANSACTION IS:

- RECORDED PERMANENTLY
- VISIBLE TO EVERYONE
- CANNOT BE CHANGED
- VERIFIED BY THE NETWORK

IN DEFI:

- USERS USE DIGITAL WALLETS
- TRANSACTIONS GO TO THE BLOCKCHAIN
- SMART CONTRACTS AUTOMATICALLY EXECUTE ACTIONS
- NO HUMAN CONTROL OR MIDDLEMEN NEEDED

SMART CONTRACT IN DIFI

- SMART CONTRACTS ARE **AUTOMATIC DIGITAL PROGRAMS**.

THEY:

- FOLLOW PREDEFINED RULES
- EXECUTE TRANSACTIONS AUTOMATICALLY
- REMOVE MANUAL PROCESSING
- REDUCE FRAUD AND ERRORS
- INCREASE SPEED AND EFFICIENCY

EXAMPLE:
IF CONDITIONS ARE MET → TRANSACTION HAPPENS AUTOMATICALLY
NO BANK APPROVAL NEEDED

TRANSACTION VALIDATION

- **TRANSACTIONS ARE VALIDATED USING CONSENSUS ALGORITHMS LIKE:**
- PROOF OF WORK (POW)
- PROOF OF STAKE (POS)
- PROCESS:
- USER SENDS TRANSACTION
- NETWORK VERIFIES IT
- CONSENSUS IS ACHIEVED
- TRANSACTION ADDED TO BLOCKCHAIN
- DATA BECOMES PERMANENT
- **THIS ENSURES:**
- SECURITY
- TRUST
- TRANSPARENCY
- DATA INTEGRITY

COMPARISON RESULTS

Parameter	Traditional Finance	DeFi
Transaction Time	Minutes to Days	Seconds
Transaction Cost	High	Low
Intermediaries	Required	Not Required
Transparency	Low	High
Accessibility	Limited	Global
Control	Centralized	Decentralized

CHALLENGES

MAJOR CHALLENGES IN DEFI:

- SCALABILITY OF BLOCKCHAIN NETWORKS
- SMART CONTRACT VULNERABILITIES
- SECURITY RISKS
- REGULATORY UNCERTAINTY
- NETWORK CONGESTION
- HIGH TRANSACTION FEES DURING PEAK USAGE

CONCLUSION

DEFI IS A POWERFUL ALTERNATIVE TO TRADITIONAL FINANCE.
IT OFFERS:

- FASTER TRANSACTIONS
- LOWER COST
- HIGH TRANSPARENCY
- GLOBAL ACCESS
- AUTOMATION THROUGH SMART CONTRACTS
- HOWEVER, FOR FULL SUCCESS, DEFI NEEDS:
- BETTER SCALABILITY SOLUTIONS
- STRONG SECURITY SYSTEMS
- SMART CONTRACT AUDITING
- CLEAR REGULATIONS
- TECHNOLOGICAL IMPROVEMENTS
- DEFI REPRESENTS THE **FUTURE OF DIGITAL FINANCE**.

THANK YOU

ANY QUESTIONS?